KHADIJA ENNAIFER

Data scientist

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PROFILE

Data Scientist with a completed first year of an MSc inData Science & Advanced Analytics at NOVA IMS, currently carrying out a thesis internship at Airbus on RAG for aviation maintenance. Hands-on experience in LLMs, NLP, OCR, machine learning, and end-to-end data pipelines. Strong in Python, SQL, and Power BI, with experience developing AI and analytics solutions from data preparation to deployment.

PROFESSIONAL EXPERIENCE

Data Science & AI Consultant BI4YOU — Tunisia June 2024 – May 2025

- Developed interactive Power BI dashboards on Snowflake usage data to monitor business KPIs, query performance and operational efficiency.
- Designed and optimized Snowflake data models, SQL queries and warehouse configurations to improve analytics performance and reduce processing times.
- Implemented automated monitoring and reporting using Snowflake stored procedures and usage-tracking data.
- Collaborated with business stakeholders to translate requirements into actionable analytics solutions.

Data Science & AI Intern Airbus — Lisbon, Portugal October 2026 – Present

- Designing and implementing a Retrieval-Augmented Generation (RAG) system that delivers relevant and traceable answers from large, heterogeneous aviation maintenance documentation.
- Comparing alternative retrieval strategies using retrieval metrics, and evaluating generated answers on relevance, faithfulness and usefulness.
- Applying a design-science approach combined with experimental evaluation
  Key skills: Python, LLMs, RAG, NLP, information retrieval

Data Science & AI Intern Welyne — Tunisia June 2025 – December 2025

- Built an end-to-end invoice processing system combining OCR (PaddleOCR, Tesseract, EasyOCR) and a locally hosted, fine-tuned LLaMA model to extract seller, buyer, total amount, currency, addresses and IBAN.
- Designed a fully on-premise pipeline (no external API) with a regex fallback layer and batch processing, ensuring data confidentiality and high accuracy on document parsing.
- Developed a Flask web application with database storage and a manual review interface; validated on real client invoices.
  Key skills: Python, LLM fine-tuning, NLP, OCR, Flask, document intelligence

End-of-Studies Internship Systra Dubai — United Arab Emirates February 2024 – June 2024

- Designed an end-to-end BI solution: data integration, data warehouse and Power BI visualizations.
- Built and deployed classification and regression models via a user-friendly web interface.
- Key skills: Python, Talend, SQL, Power BI

Business Intelligence Intern Systra Dubai — United Arab Emirates May 2023 – August 2023

- Performed a full ETL pipeline: data collection, extraction, cleaning and transformation.
- Created interactive dashboards in Power BI using DAX.
- Developed a web interface using HTML, CSS, JavaScript and web frameworks.

Volunteer Instructor - English and Computer Science Seif Center — Tunisia 2023 – 2025

EDUCATION

MSc Data Science & Advanced Analytics (Major in Data Science) 2025 - 2027
NOVAIMS - Portugal

Bachelor in Business Intelligence 2021 - 2025
ESEN La Manouba - Tunisia

Baccalaureate (Tunisian + French) 2017 - 2021
Enasser High School - Tunisia

PROJECTS

Painting Classification (WikiArt) - Deep Learning Project, NOVA IMS - 2026

- Built a deep learning workflow to classify paintings by artist, from preprocessing and baseline to transfer learning with DenseNet-121, EfficientNet and Vision Transformer (ViT).
- Developed reusable evaluation utilities (training curves, confusion matrix, classification report) packaged as a Python module.
  Key skills: Python, TensorFlow, Computer Vision, scikit-learn

Airline Customer Segmentation, Best project (20/20) - 2025

- Appliedclusteringtechniques(K-Means,SOM, Autoencoders,HDBSCAN, GMM) to segment an airline's customer base into 6 behavioral & value-based profiles.
- Delivered personalized marketing strategies per segment to improve retention and loyalty.
- Built a Streamlit dashboard to visualize the results.

ACHIEVEMENTS

Crustacean Age Prediction Hackathon – Gradient Boosting model(2024)
El Maken Hackathon – KNN classification + BI dashboard (2023)
Mathematics Competition – Represented Tunisia (2017)

SKILLS

- **Key Skills:** Python, LLM fine-tuning, RAG, NLP, OCR
- **Data & Business Intelligence:** Power BI / DAX, SQL / Oracle, Snowflake, MongoDB, ETL / Data Mining
- **AI / Machine Learning:** PyTorch, Scikit-learn, TensorFlow, Deep Learning, Computer Vision
- **Development:** HTML / CSS / JavaScript, Flask, Git

LANGUAGES

- Arabic: C2
- French: C2
- English: C1
- Portuguese: A1